

ProMaize: which lines get the ground

SyDAg 2026 · Bayer Genomic Prediction track · Alexander Baena, Sofia Gerena, Alen Lizarazo · [promaize.stipe.app](#)

The decision. January 2008: plots are cut, and 15,962 new lines from 157 biparental families wait for their first testcross season. Every family is new; 24 have neither parent on record. A line that is not tested can never be advanced.

The insight. This program tests each family once, so a model that learns “which families were good” has nothing to say about next year’s. ProMaize splits the question the way a breeder does:

- **the family’s mean GCA** from what its parents passed on (the family’s mean genotype, which gets backcrosses right), recent years weighted up;
- **which siblings** from which parental segments each line inherited (a sibling-difference model, where tester, trial and family effects cancel);
- the **tester’s effect removed** as a BLUP, so lines are not credited for their tester;
- yield, moisture, maturity and lodging priced into **\$/acre** at the breeder’s prices.

Evidence, from seasons the model never saw (each year’s families predicted from earlier years only):

	ProMaize	standard GBLUP	pedigree BLUP
forward r, mean of 2003-2008	0.22	0.15	0.06
forward r, 2008 (decision year)	0.13	0.07	0.05
seasons won against standard GBLUP	6 of 6		
realised \$/acre over random, plant 30%, 6-season mean	\$8.8	\$5.5	
real top 10% kept, plant 30% (random: 30%)	41%	36%	

- 2008 gain over standard GBLUP: 0.03 to 0.11 in r (95%, resampling families); 6 of 6 seasons, sign test $p = 0.016$.
- 90% bands held **90.2%** of real 2008 results. Plot noise caps any predictor near $r = 0.68$.
- Standard GBLUP needed **1,277 more lines** (about 8,900 plots) in 2008 to keep the same winners.
- Location-specific response is not predictable here ($r = 0.03$ on 82,790 held-out plots): broad-acre GCA is the right target at this stage.

Recommendation for 2008. Plant the same share of every family and let markers choose the siblings (docs/advance_2008.csv, 4,789 lines, all 157 families). With pedigree this thin, the family call is weak: in the two earlier thin seasons (2003, 2007) the even split cost \$2.02 and \$0.33/acre for about twice the families, against \$4.5-7.5 with a well-recorded pedigree; in 2008 it cost \$0.08. With a well-recorded pedigree, rank every line by \$/acre instead (docs/advance_2008_ranked.csv).

Rigor. Settings chosen on 2005-2007; 2008 held out. The first frozen model scored 0.120 on 2008; three later changes, each chosen on 2005-2007 and applied to every year, are disclosed. A leaky random k-fold reads 0.56 on the same lines and is shown only for contrast.

Run it. `python scripts/make_fixture.py && python scripts/build_data.py --judge`: the whole pipeline on a synthetic program in the real file layout, 10 seconds, four packages. Full data: about five minutes on a laptop; training uses per-season sufficient statistics, so a new year is an update.